

Initial Project Plan (week 10, submission date: 5 June 2026)

Group Name		AIR SELANGOR	
Members			
Name		Email	Phone number
MUHAMMAD ASLAM ROFI BIN MOHD RASU		230405@student.upm.edu.my	0148077505
AQIL QAYYIM BIN ABDUL KHALID		228096@student.upm.edu.my	0173117966
HERMAWAN RYAN IQBAL		226745@student.upm.edu.my	01140662236
ABDULLAH ALI IMRAN BIN IFFLI EFENDY		sa00076@student.upm.edu.my	0108716138
Problem scenario description	Northern Malaysia, specifically the "rice bowl" states of Kedah and Perlis, is experiencing a severe and prolonged drought. Central reservoir levels are critically low, and local farming districts face an imminent risk of widespread crop failure. The state government must deploy a limited fleet of water distribution trucks to multiple regions		
Why it is important	This crisis directly threatens national food security and the economic survival of rural farming communities. Water is strictly limited; a suboptimal distribution means wasting resources on low-priority or inaccessible areas, resulting in the avoidable death of highly vulnerable and critical farmlands.		
Problem specification	Data Types: A list of Node objects representing farming districts, containing variables for waterRequired, priorityScore, waterAllocated, and distanceFromReservoir. Constraints: The sum of water allocated cannot exceed the totalReservoirCapacity. Water may be continuous (liters) or discrete (whole trucks). Objective Function: Maximize the overall crop survival rate, defined as the sum of Water Allocated x Priority Score, while identifying the shortest physical transport routes		
Potential solutions	1. Greedy Algorithm (Fractional Knapsack): Allocates continuous water rapidly based on a priority-to-need ratio. 2. Dynamic Programming (0/1 Knapsack): Maximizes priority scores for discrete, indivisible water truck deliveries. 3. Dijkstra's Algorithm (Graph Routing): Calculates the most cost-effective and shortest physical transport routes from the central reservoir to the districts.		
Sketch (framework, flow, interface)	Framework: Model-View-Controller (MVC) architecture separating pure algorithmic logic from the UI. Flow: The user inputs the total available water capacity -> selects an algorithm -> the system computes the optimal distribution and tracks execution time -> the UI updates with the results. Interface: A JavaFX desktop application featuring a top control panel with three execution buttons (Greedy, DP, Dijkstra) and a monospaced text-area dashboard that displays a clear, tabular summary of regions, water needed, water allocated, and execution time in nanoseconds.		

Project Proposal Refinement (week 11, submission date: 12 June 2026)

Group Name	AIR SELANGOR		
Members			
	Name	Role	
	MUHAMMAD ASLAM ROFI BIN MOHD RASU	Programmer	
	AQIL QAYYIM BIN ABDUL KHALID	Designer for JavaFX	
	HERMAWAN RYAN IQBAL	Paperwork and slides	
	ABDULLAH ALI IMRAN BIN IFFLI EFENDY	Researcher and data analyst	
Problem statement	The agricultural sector in Northern Malaysia (Kedah and Perlis) is facing a critical water shortage due to a prolonged drought. With a limited centralized water supply, the government struggles to distribute water efficiently to various farming districts to prevent catastrophic crop failure and ensure food security.		
Objectives	1. To design an algorithmic model that optimally allocates a limited water supply across multiple farming districts. 2. To maximize the overall crop survival rate based on regional priority scores. 3.To calculate the shortest and most cost-effective physical routing for water distribution trucks. 4. To analyze and compare the time complexity, efficiency, and real-world viability of Greedy, Dynamic Programming, and Dijkstra's algorithms		
Expected output	A JavaFX desktop application that accepts total water capacity as an input, executes three distinct distribution algorithms, and outputs a tabular dashboard showing the allocated water per region, total priority score saved, and the exact nanosecond execution time for comparative analysis.		
Problem scenario description	Northern Malaysia, specifically the "rice bowl" states of Kedah and Perlis, is experiencing a severe drought. The central reservoir is critically low. The government must deploy water trucks to multiple regions, but the supply is strictly insufficient to meet all demands simultaneously.		
Why it is important	This crisis directly threatens national food security and the economic survival of rural farming communities. Water is strictly limited; a suboptimal distribution means wasting resources on low-priority or physically inaccessible areas, resulting in the avoidable death of highly vulnerable and critical farmlands.		
Problem specification	Data Model: Node objects representing districts with properties: waterRequired, priorityScore, waterAllocated, and distanceFromReservoir. Constraints: sum of Water Allocated x Total Reservoir Capacity. Water is treated as either continuous (liters) or discrete units (whole trucks). Objective Function: Maximize the survival score, defined as sum of Water Allocated x Priority Score, while minimizing transport distance.		
Potential solutions	1. Greedy Algorithm (Fractional Knapsack): Provides fast, $O(N \log N)$ continuous water allocation based on a priority-to-need ratio. 2. Dynamic Programming (0/1 Knapsack): Provides the absolute optimal allocation for discrete, indivisible units, though it requires heavy $O(NW)$ computational time. 3. Dijkstra's Algorithm (Graph Routing): Calculates the shortest physical transport routes from the reservoir to the districts in $O(V^2)$ time.		
Sketch (framework, flow, interface)	Framework: Model-View-Controller (MVC) architecture separating pure algorithmic logic from the UI.		

	Flow: The user inputs the total available water capacity -> selects an algorithm -> the system computes the optimal distribution and tracks execution time -> the UI updates with the results. Interface: A JavaFX desktop application featuring a top control panel with three execution buttons (Greedy, DP, Dijkstra) and a monospaced text-area dashboard that displays a clear, tabular summary of regions, water needed, water allocated, and execution time in nanoseconds.													
Methodology	<table><tr><th>Milestone</th><th>Time</th></tr><tr><td>Find a disaster in Malaysia and make a scenario</td><td>Week 10</td></tr><tr><td>Find the solution and researching the algorithms. Start doing the barebone of code and do testing.</td><td>Week 11</td></tr><tr><td>Debugging the coding and turning normal java into JavaFX</td><td>Week 12</td></tr><tr><td>Analysis of all the algorithm and decide the best one</td><td>Week 13</td></tr><tr><td>Presentation in class</td><td>Week 14</td></tr></table>		Milestone	Time	Find a disaster in Malaysia and make a scenario	Week 10	Find the solution and researching the algorithms. Start doing the barebone of code and do testing.	Week 11	Debugging the coding and turning normal java into JavaFX	Week 12	Analysis of all the algorithm and decide the best one	Week 13	Presentation in class	Week 14
Milestone	Time													
Find a disaster in Malaysia and make a scenario	Week 10													
Find the solution and researching the algorithms. Start doing the barebone of code and do testing.	Week 11													
Debugging the coding and turning normal java into JavaFX	Week 12													
Analysis of all the algorithm and decide the best one	Week 13													
Presentation in class	Week 14													